

Multiple Choice (10 marks)

1. 1. _____ framework made cracking of vulnerabilities easy like point and click.
 - (a) .Net
 - (b) Metasploit
 - (c) Zeus
 - (d) Ettercap
2. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Dark web
 - (d) Surface web
3. Let $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x^{**2}]$. The value of $f(4)$ is
 - (a) 53
 - (b) 29
 - (c) 50
 - (d) 100
4. In Message Confidentiality, the transmitted message must make sense to only intended
 - (a) Receiver
 - (b) Sender
 - (c) Modulator
 - (d) Translator
5. When a wireless user authenticates to any AP, both of them go in the course of four-step authentication progression which is called _____
 - (a) AP-handshaking
 - (b) 4-way handshake
 - (c) 4-way connection
 - (d) wireless handshaking

True / False (10 marks)

Answer True or False

6. True or False: The sub key length at each round of DES is 48 bits, which is used for key expansion during the encryption process.
7. True or False: Interrupts are used in place of data channels to transfer data between devices without CPU intervention.
8. True or False: The length of the message digest produced by the MD 5 hashing algorithm is 160 bits.
9. True or False: The digest created by a hash function is typically referred to as a Modification Detection Code (MDC).
10. True or False: The priority queuing job-scheduling policy guarantees that all jobs receive service without indefinite waiting, making it starvation-free.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to split a string at lowercase letters.
12. Write a python function to find element at a given index after number of rotations.

End of Paper